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BSIT 303I

1. Csv File

```
import pandas as pd
import matplotlib.pyplot as plt

# Read data from CSV file
df = pd.read_csv("sales_data.csv")

# Display the data
print(df)

# Create a line chart
plt.plot(df["Month"], df["Sales"],
         marker="o", label="Sales", color="blue")

plt.plot(df["Month"], df["Expenses"],
         marker="s", label="Expenses", color="red")

# Add chart information
plt.title("Monthly Sales and Expenses")
plt.xlabel("Month")
plt.ylabel("Amount ($)")
plt.legend()
plt.grid(True)

# Display chart
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
      Month  Sales  Expenses
0   January  15000      8000
1  February  18000      9000
2     March  22000     11000
3     April  25000     12000
4       May  30000     15000
5      June  28000     14000
```



2. Website CSV

```python
import pandas as pd
import matplotlib.pyplot as plt

# URL of online CSV dataset
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/flights.csv"

# Read data directly from website
df = pd.read_csv(url)

# Convert year and month into dates
df["Date"] = pd.to_datetime(
    df["year"].astype(str) + "-" + df["month"],
    format="%Y-%B"
)

# Display first 10 records
print(df.head(10))

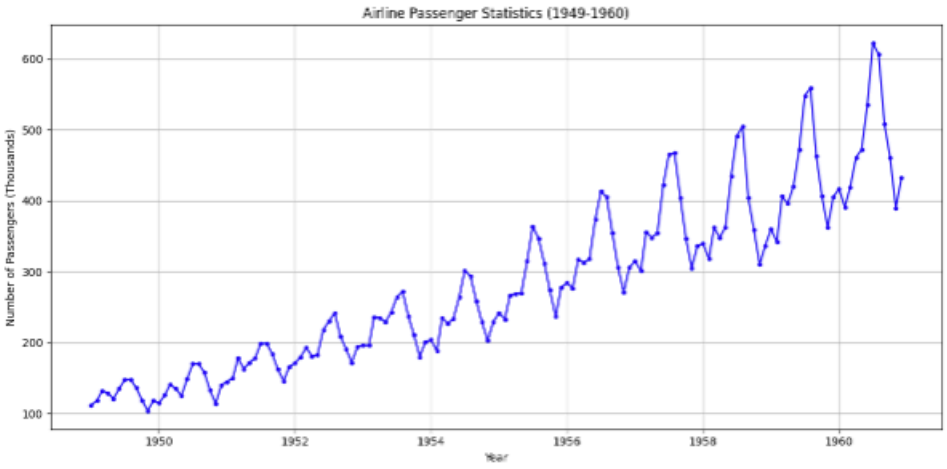
# Create line chart
plt.figure(figsize=(12, 6))

plt.plot(
    df["Date"],
    df["passengers"],
    color="blue",
    marker="o",
    markersize=3,
    linewidth=1.5
)

# Chart title and labels
plt.title("Airline Passenger Statistics (1949-1960)")
plt.xlabel("Year")
plt.ylabel("Number of Passengers (Thousands)")

plt.grid(True)
plt.tight_layout()
plt.show()
```

```
   year     month  passengers       Date
0  1949   January         112 1949-01-01
1  1949  February         118 1949-02-01
2  1949     March         132 1949-03-01
3  1949     April         129 1949-04-01
4  1949       May         121 1949-05-01
5  1949      June         135 1949-06-01
6  1949      July         148 1949-07-01
7  1949    August         148 1949-08-01
8  1949 September         136 1949-09-01
9  1949   October         119 1949-10-01
```



3. Excel File

```
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files

# Read CSV file
df = pd.read_csv("company_data.csv")

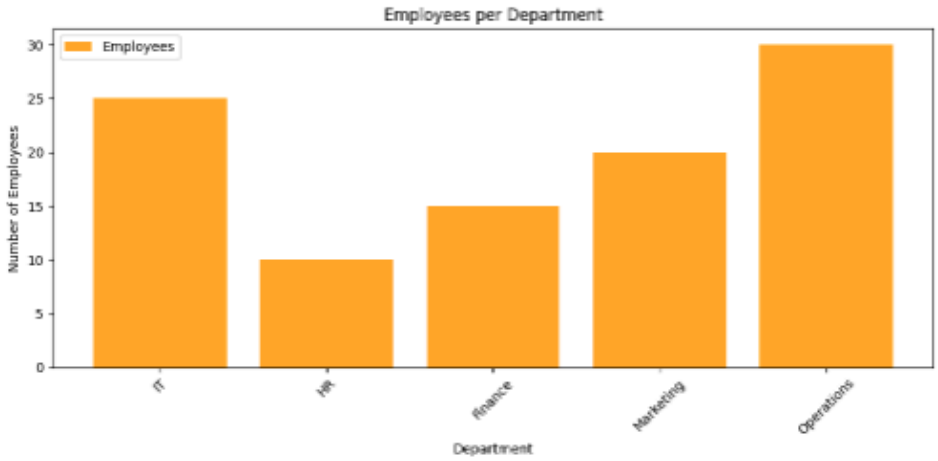
# Display records
print(df.head())

# Create bar chart
plt.figure(figsize=(10, 5))

plt.bar(df["Department"], df["Employees"],
        color="orange", label="Employees")

plt.title("Employees per Department")
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
   Department  Employees
0          IT         25
1          HR         10
2     Finance         15
3   Marketing         20
4  Operations         30
```



3 Different types of Approaches:

The three different approaches are using a CSV file, getting data directly from a website, and using an Excel file. The CSV approach uses `pd.read_csv()` to read data from a saved file, while the website approach uses the same function to get data from an online source. On the other hand, the Excel approach uses `pd.read_excel()` to read data from an Excel file. The main difference between these approaches is where the data comes from, but all of them help us create charts using Pandas and Matplotlib to make information easier to understand and compare.